Artificial Intelligence to Diagnose Ischemic Stroke and Identify Large Vessel Occlusions: A Systematic Review

Comprehensive Research Paper Profile and Analytical Summary

Citation & Study Overview

Attribute	Specification
Authors	Nathan M. Murray, Mathias Unberath, Gregory D. Hager, Ferdinand K. Hui
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Design Standard	Systematic Review (PRISMA-Compliant)
Primary Domain	Neurointerventional Radiology / Deep Learning Diagnostics

Background & Rationale

Acute ischemic stroke (AIS) secondary to large vessel occlusion (LVO) is a medical emergency requiring rapid intervention. Endovascular thrombectomy (EVT) drastically improves functional .("outcomes, yet its therapeutic efficacy is intensely time-dependent ("time is brain

- **Clinical Bottlenecks:** Manual interpretation of emergency neuroimaging often introduces latency, particularly in community or non-academic settings where specialized neuroradiology coverage is not available 24/7.
- **Core Objective:** To systematically review, evaluate, and benchmark machine learning (ML) and deep learning (DL) models designed for automated detection of ischemic stroke, quantification of tissue viability via the Alberta Stroke Program Early CT Score (ASPECTS), and identification of LVO across multimodal imaging protocols.

Methodology & Search Protocol

The systematic review followed standard Preferred Reporting Items for Systematic Reviews and .Meta-Analyses (PRISMA) guidelines

- **Search Databases:** Comprehensive searches executed across PubMed, Scopus, Embase, and Web of Science.
- **Inclusion Criteria:**
    - Peer-reviewed primary research articles implementing AI, machine learning, or deep learning algorithms directly on diagnostic neuroimaging.
    - Imaging modalities including non-contrast computed tomography (NCCT), CT angiography (CTA), CT perfusion (CTP), and magnetic resonance imaging (MRI).
    - Target outputs focused on detection of acute ischemia, measurement of salvageable versus infarcted tissue, or localization of arterial occlusions.
- **Exclusion Criteria:** Editorials, conference abstracts lacking complete methodology, reviews, and studies focusing strictly on non-imaging electronic health record data.

Evaluated Technologies & Imaging Modalities

The systematic analysis categorizes state-of-the-art neuroimaging AI applications across three :primary imaging workflows

Modality	Computational Architectures	Primary Clinical Tasks
Non-Contrast CT (NCCT)	2D/3D Convolutional Neural Networks (CNNs), DenseNets	Detection of subtle early ischemic changes; quantitative assessment of tissue hypodensity; automated regional ASPECTS scoring (0–10 scale).
CT Angiography (CTA)	Deep 3D CNNs, Vessel Segmentation Networks	Automated detection of arterial cutoffs across the internal carotid artery (ICA), middle cerebral artery (MCA-M1 and M2 segments), and basilar artery.
CT Perfusion (CTP) & MRI	Thresholding ML, Fully Convolutional Networks (FCNs)	Volumetric quantification of core infarct (cerebral blood flow reduction) versus penumbra

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		(hypoperfused tissue defined by Tmax > 6s delays).

Key Findings & Diagnostic Performance

- **LVO Diagnostic Accuracy:** Advanced deep learning models applied to CTA demonstrated high diagnostic performance, routinely achieving sensitivities and specificities between 85% and 95% for proximal anterior circulation occlusions (ICA and MCA-M1).
- **ASPECTS Scoring Concordance:** AI-driven automated ASPECTS systems exhibited scoring agreement and intraclass correlation coefficients comparable to expert fellowship-trained neuroradiologists, significantly outperforming non-expert emergency clinicians and general radiologists.
- **Triage Speed and Processing Latency:** End-to-end algorithmic inference times were consistently under 3 to 5 minutes from image reconstruction, facilitating instantaneous automated notifications and parallel workflow activations across comprehensive stroke teams.

Limitations & Future Directions

While the performance metrics across initial cohorts are promising, several critical translational :barriers were highlighted

- **Single-Center Retrospective Bias:** The majority of evaluated literature relied on retrospective datasets collected from single academic centers. This limits algorithmic generalizability across heterogeneous CT scanners, varied slice thicknesses, and differing contrast injection protocols.
- **Distal and Posterior Detection Gaps:** Current deep learning models exhibit high sensitivity for proximal anterior circulation lesions (ICA and M1-MCA), but show a substantial drop in performance when detecting distal branch occlusions (M2/M3-MCA segments) and posterior circulation infarctions (vertebral and basilar arteries).
- **Lack of Prospective Outcome Evidence:** Most available studies focused solely on standalone diagnostic accuracy (AUC, sensitivity, specificity) in offline evaluations. Prospective randomized controlled trials evaluating the concrete impact of automated AI alerting on door-to-needle time, door-to-groin puncture time, and long-term 90-day modified Rankin Scale (mRS) outcomes remain necessary.